Pushpam Raj Satyarthi

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CSE undergraduate (CGPA 9.17) who goes deep on backend and data systems, then ships whatever else the product needs: APIs, ETL pipelines, ML models, frontends, even a native Mac app. Production backend experience on a live AI product. Targeting SDE roles.

EXPERIENCE

Motherson Technology Services — May – Jun 2026
Software Engineering Intern, AIML team — Hyderabad, India

- Designed and shipped a hybrid persistence layer for an AI legal-research platform: SQLite for conversation metadata, Azure Blob Storage for full message payloads, end to end across a FastAPI backend and three React/TypeScript modules.
- Wrote a 316-line FastAPI module from scratch with 9 REST endpoints (save, restore, search, archive) keeping Blob writes and SQLite metadata consistent, with storage isolated per product mode so sessions never leak across modules.
- Audited the codebase and surfaced 14+ bugs, including broken mode isolation in all three modules; logged them in a 27-entry tracker and worked through root causes with the team.

PROJECTS

Threat Detection Engine · Python, XGBoost, SHAP, FastAPI, React/TS, Playwright, NetworkX — live code

- Fake-account and botnet detector for SIH-1775 (ITBP / MHA): two platform-specific XGBoost classifiers on 101,000+ labelled accounts, 96.87% accuracy on Instagram/Facebook and 90.04% on X, with SHAP reasons in plain English.
- Found a mislabelled 50,000-row dataset capping X accuracy at 67%; removing it and splitting feature pipelines (13 vs 7 features) added 23 points.
- Live profile scans with headless Chromium, a NetworkX graph of coordinated accounts, a one-way case workflow and one-click PDF case files.

Orbitune · JavaScript, Three.js, Node.js/Express, IndexedDB, Spotify API — live code

- Turns a Spotify listening history into a 3D universe: an in-browser ETL parses full streaming exports (100 MB zips never leave the browser); my own map holds 3,363 artists and 51,772 plays.
- Louvain community detection on an artist-similarity graph, a 3D UMAP layout and TF-IDF cluster names; OAuth tokens kept in an AES-256-GCM encrypted cookie, no database.

Sluice · Python, TOML, SQLite, zero dependencies — code

- Data pipeline runner that refuses to fail quietly: checks between stages stop the run and name the exact row, column and value at fault. Run as an open-source practice ground for contributors.

Global Banking ETL · Python, Pandas, NumPy, SQL — code

- Fault-tolerant multi-source ETL into a SQL warehouse; vectorised normalisation cut data redundancy 30%, incremental delta loads replaced full refreshes, and pre-load checks block schema mismatches.

Scrollo · Swift, CGEventTap, Accessibility API, AVAudioEngine — code

- Free, open-source macOS menu-bar app that clicks and taps the trackpad as you scroll, staying silent at page edges by reading scroll position through the Accessibility API; idles near 0% CPU.

SKILLS

Languages	Python, SQL, TypeScript, JavaScript, C, C++, Java, Swift
Backend & data	FastAPI, Node.js, Express, ETL design, incremental loads, schema normalisation, data warehousing
Databases	PostgreSQL, MySQL, SQLite, MongoDB (raw SQL, stored procedures, query optimisation)
ML	XGBoost, scikit-learn, Pandas, NumPy, SHAP, SentenceTransformers
Cloud & tools	Azure Blob Storage, Vercel, Git, Linux shell, Playwright, Valgrind, GDB, Make

EDUCATION & CERTIFICATION

GITAM University, Visakhapatnam — 2024 – 2028
B.Tech, Computer Science and Engineering — CGPA 9.17

IBM Data Engineering Professional Certificate (Coursera, 16 courses): Airflow, Kafka, Spark, Hadoop, data warehousing, BI.

LEADERSHIP & ACHIEVEMENTS

President, IEEE-CS GITAM Vizag · Domain Lead, Open Source, GitHub Community GITAM Vizag

Smart India Hackathon internal-round qualifier, 2026 (SIH-1775) and 2025 (TripSync, built in 24 hours) · Gitathon: full backend in 24 hours